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HANDELSHØJSKOLEN

LOGISTIC REGRESSION

Classification Model

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Agenda

Why Logistic Regression (LR)?

- Generative vs. Discriminative Classifier
- Linear vs. Logistic Regression

Introduction to LR

- Log Odds
- Sigmoid Function
- MLE

Implementation

- LR in scikit-learn
- Evaluation

Key Takeaways

- Assumption
- Limitation

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Generative vs. Discriminative Classifier

Generative Classifiers	Discriminative Classifiers
Tries to model class - what are the features of the class.	Learn what features are most useful to distinguish between the various classes.
Learns the joint probability distribution p(x,y) of label x and y. It predicts the conditional probability and then picking a most likely label.	Learns the conditional probability distribution p(y|x).
Example: Naive Bayes	Example: Logistic Regression
$\hat{c} = \underset{c \in C}{\operatorname{argmax}} \overbrace{P(d\|c)}^{\text{likelihood}} \overbrace{P(c)}^{\text{prior}}$	$\hat{c} = \underset{c \in C}{\operatorname{argmax}} \overbrace{P(c\|d)}^{\text{posterior}}$

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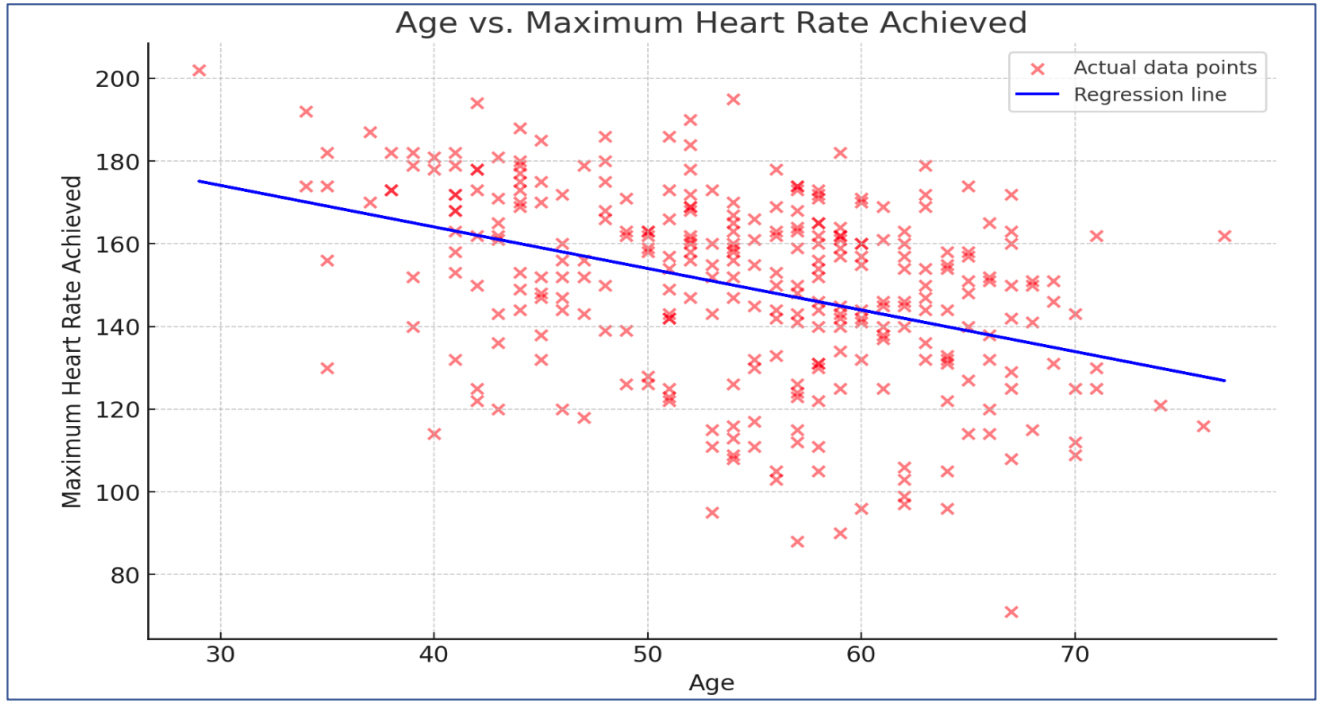
Regression Model

Goal - Find the best-fitting model showing the relationship between predictors and outcome variables.



A statistical method used to understand the relationship between

- A dependent variable (Outcome)
- One or more independent variables (Predictors)

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Linear Regression

Alice and her Cookie

Predicts a continuous outcome based on the linear relationship between predictors(x) and outcome (y).

$\hat{y}$ =mx+b

How the number of advertisements (x) each week affects weekly cookie sales (y)?

Weeks	No. of Ads	Sold Cookies
Week 1	2 ads	50
Week 2	3 ads	60
Week 3	4 ads	75

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Logistic Regression

Alice and her Cookie

Models the probability that each input belongs to a particular class/category

- based on linear combinations of input features passed through a logistic function.

$$\hat{y} = \frac{1}{1+e^{-z}}$$

Predict whether a week sale will be ”high" (1) or "low " (0) based on the number of ads.

Assume weeks with more than 60 cookies sold as ‘high’.

Weeks	No. of Ads	Sold Cookies	Sales category
Week 1	2 ads	50	low
Week 2	3 ads	60	low
Week 3	4 ads	75	high

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Why do researchers still use Logistic Regression despite advanced ML models?

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Introduction to Logistic Regression

A regression model used for predicting (binary) outcomes based on predictor variables.

- Provides the probability score reflecting the likelihood of an event's occurrence.

Input observation:

A vector

$X = [x_1 \ldots x_n]$

Weights: One per feature

$w = [w_1 \ldots w_n]$

Linear Combination

$z = b + \sum_{i=1}^{n} w_i . x_i$

Apply Sigmoid function on z to get the value in (0,1) range

Output: A predicted class

$\hat{y} = \{0,1\}$

Logistic Regression for one observation x (consist of n features)

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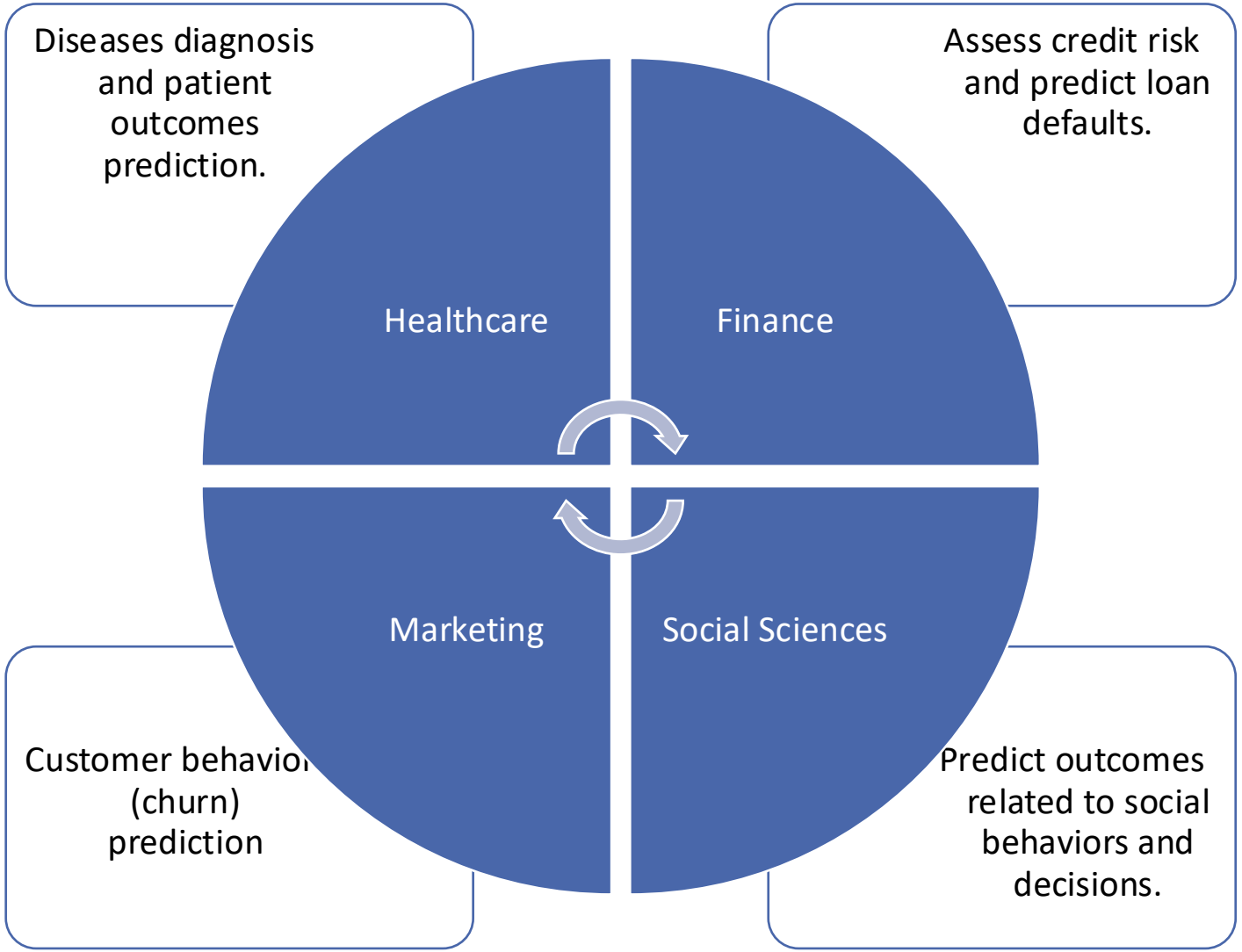
Real-world Applications



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How familiar are you with Logistic Regression?

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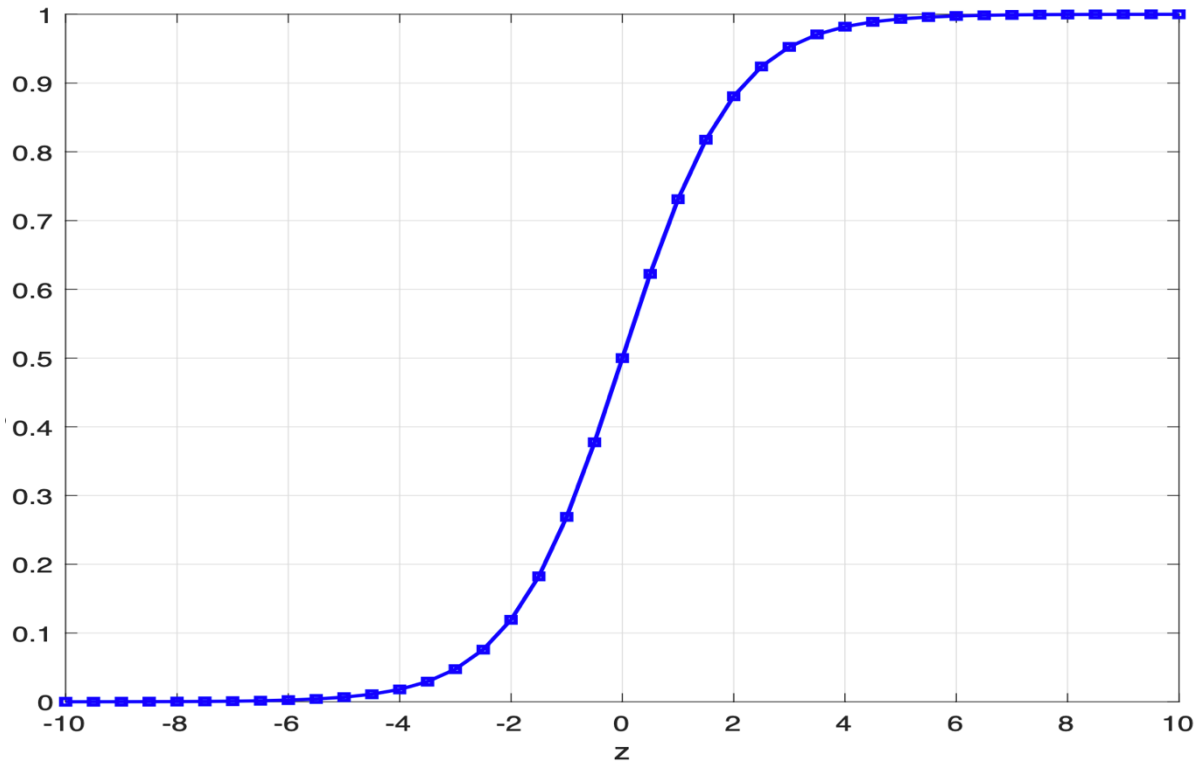
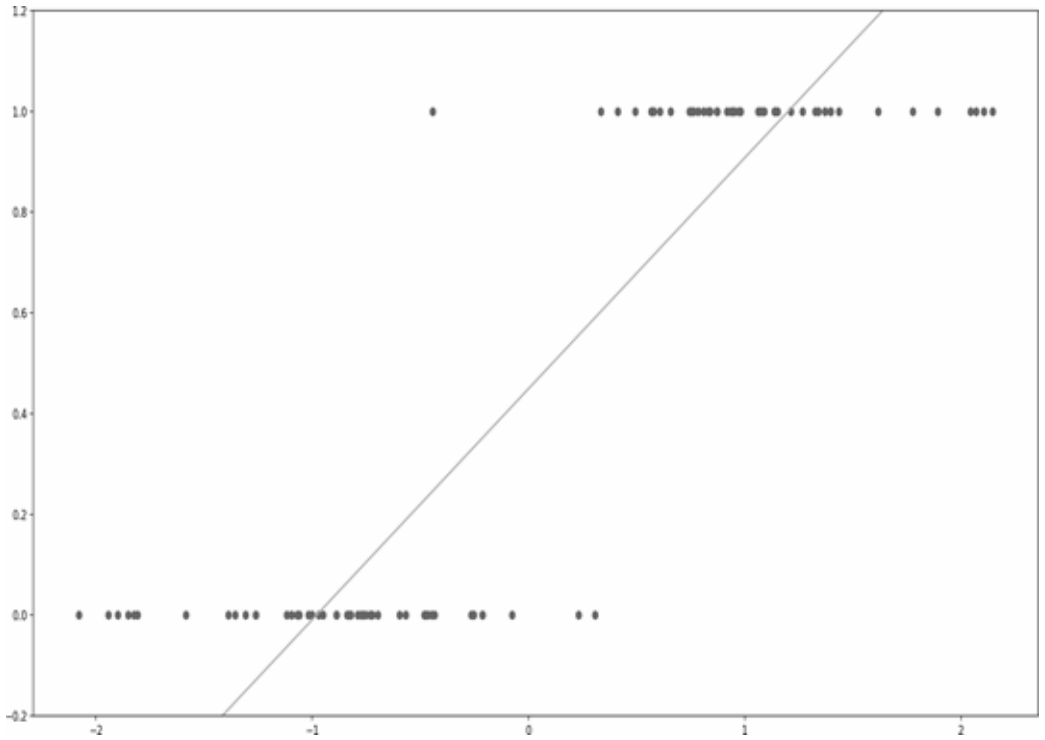
The Heart of LR: The Sigmoid Function

"Converting linear output to probabilities"





$$\sigma(z) = \frac{1}{1+e^{-z}}$$

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Log-Odds in Logistic Regression

- Odds $= \frac{P}{1-P}$ $\longrightarrow$ Log–Odds $= \log\left(\frac{P}{1-P}\right)$
  - P -> probability of the event occurring
  - (1-P) -> probability of the event not occurring
- Log-odds of the probability of the positive class (Y=1) can be modelled as a linear combination of the input features (X=$[x_1 \dots x_n]$)

$$\log\left(\frac{P(Y=1|X)}{1-P(Y=1|X)}\right) = b + w_1x_1 \dots w_nx_n$$

  - P(Y=1|X) - Probability of the positive class given the input features X
  - $w = [b, w_1 \dots w_n]$ - coefficients (weights)

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Log-Odds and Sigmoid

Question

How to convert the log-odds back into a probability?

Answer

Use sigmoid function

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$P(Y = 1|X) = \sigma(b + w_1x_1 \dots w_nx_n) = \frac{1}{1 + e^{-(b+w_1x_1 \dots w_nx_n)}}$$

Model learns the coefficients (b, w) that predict the log-odds of the positive class.

Using sigmoid function, these predictions are transformed into probabilities (0,1).

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Fitting a Model with MLE

Maximum Likelihood Estimation(MLE)

- A method used to find the model parameters that most likely explain the observed outcomes.

Purpose

- Ensure that model's predicted probabilities closely match with the actual (binary) outcomes.
- Optimize the model for accurate predictions and classification.

Why MLE?

- Able to directly work with probabilities and handle the non-linear nature of the logistic function
- Helps in adjusting the parameters so that predicted result reflect actual outcomes as closely as possible.

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Wait, where did the w's come from?

We know the correct label **y** (0 or 1) for each feature.

- But what the system produces is an estimate, $\hat{y}$

We want to set the value of w and b to minimize the distance between $\hat{y}$ and y

Need of a distance estimator: a loss or a cost function

- Cross-entropy loss

Need of an optimization algorithm to update w and b to minimize the loss.

- Stochastic gradient descent

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Process of finding the best model parameters

Define the Logistic Regression Model

→

Construct the likelihood function and convert it into log-likelihood

→

Start with initial parameter values and iteratively update them

- until log-likelihood converges to a maximum.

→

Use numerical methods (gradient descent) to find the optimal parameter values

Importance of MLE in Ensuring Model Accuracy

Optimal Parameter Estimation

- Finds the most probable values for the observed data

Consistency

- With more data, MLE estimates get closer to the true values.

Efficiency

- Lowest variance among all unbiased estimators

Foundation for Model Comparison

- Provides a basis for comparing different models through likelihood-based criteria

Implementing Logistic Regression in scikit-learn

Data Preparation

- Data Cleaning, Normalizing or standardizing numerical data
- Encoding categorical variables: convert categorical data into a numerical format

Feature selection (Predictors)

- Choose the most significant features from the dataset

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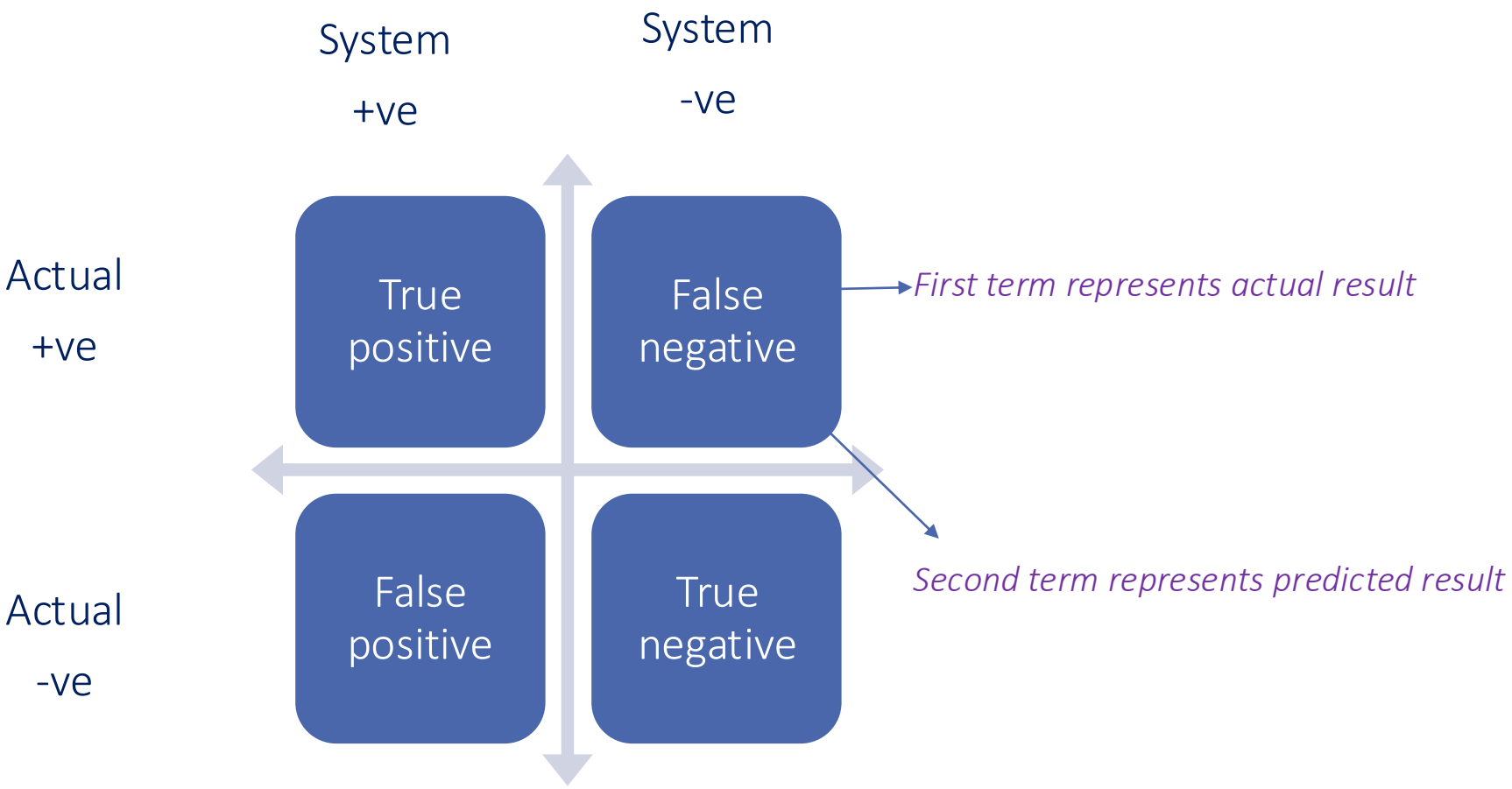
Implementing Logistic Regression in scikit-learn

Data preparation and Feature selection → Train the LR Model → Test the LR Model → Model evaluation

To see step by step python Implementation click here

Evaluation: Confusion Matrix



Precision, Recall, F-measure

Precision

$$\frac{\text{True positive}}{\text{True positive} + \text{False positive}}$$

Recall

$$\frac{\text{True positive}}{\text{True positive} + \text{False Negative}}$$

F_1-measure

$$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

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Assumption of Logistic Regression

A linear relationship between predictor variables (features) and the log odds of the outcome.

Requires that predictor variables are not highly correlated.

Observations must be independent.

Limitation of Logistic Regression

May not effectively capture complex non-linear relationships.

With many predictors or complex models, there's a risk of overfitting to the train data, leading to poor generalization to new data.

Needs a sufficiently large sample size for stable and reliable coefficient estimates.

Key Takeaways

- Primarily used for interpretable Classification.
- Odds Ratio tells how predictor variables affect the probability of the positive outcome.
- Uses the logistic (sigmoid) function to make predictions lies in (0, 1).
- Coefficients are estimated using MLE, focusing on finding the model that maximizes the likelihood of the observed data.

How Can I further help you?

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